

ALAMEDA, CA, USA

WMO: 994033

Lat: 37.772N

Lon: 122.298W

Elev: 7

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 40.6	(c) 42.6	(d) 23.8	(e) 18.0	(f) 49.9	(g) 28.7	(h) 22.7	(i) 53.6	(j) 24.6	(k) 52.5	(l) 20.8	(m) 52.5	(n) 4.9	(o) 350	(p) 0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 9	(b) 12.2	(c) 81.4	(d) 64.1	(e) 77.3	(f) 63.0	(g) 73.9	(h) 62.0	(i) 66.5	(j) 78.6	(k) 65.2	(l) 76.3	(m) 64.0	(n) 73.8	(o) 7.0	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 62.4	(b) 84.4	(c) 68.9	(d) 61.0	(e) 80.0	(f) 67.9	(g) 59.8	(h) 76.6	(i) 67.3	(j) 31.0	(k) 78.7	(l) 30.0	(m) 76.5	(n) 29.1	(o) 73.6	(p) 75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 20.1	(b) 17.5	(c) 15.4	(e) 37.1	(f) 89.5	(g) 3.5	(h) 5.1	(i) 34.6	(j) 93.1	(k) 32.5	(l) 96.1	(m) 30.5	(n) 99.0	(o) 28.0	(p) 102.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	58.6	51.8	53.8	55.8	57.9	59.2	61.8	62.9	63.6	64.6	63.1	57.3	51.6	(6)
(7)		DBStd	6.12	3.90	4.38	4.12	4.30	4.12	4.35	3.90	3.81	4.09	4.59	4.21	4.27	(7)
(8)		HDD50	76	27	13	3	0	0	0	0	0	0	0	2	31	(8)
(9)		HDD65	2484	410	314	285	218	188	116	88	72	53	91	233	416	(9)
(10)		CDD50	3233	81	120	184	238	287	355	401	420	439	407	221	80	(10)
(11)		CDD65	169	0	0	1	6	10	22	24	28	43	33	2	0	(11)
(12)		CDH74	757	0	1	4	33	54	141	101	105	160	152	6	0	(12)
(13)		CDH80	176	0	0	0	6	10	46	25	23	31	35	0	0	(13)
(14)	Wind	WSAvg	6.7	5.4	6.5	7.1	7.7	8.0	7.6	7.8	7.4	6.5	5.5	5.2	5.4	(14)
(15)	Precipitation	PrecAvg	17.9	3.7	2.8	2.8	1.2	0.2	0.1	0.0	0.2	1.3	2.7	2.6		(15)
(16)		PrecMax	35.5	9.1	7.2	8.0	5.4	2.4	1.2	0.7	0.9	1.5	8.2	7.9	6.1	(16)
(17)		PrecMin	7.2	0.3	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	(17)
(18)		PrecStd	6.9	2.6	2.1	2.0	1.3	0.4	0.3	0.2	0.2	0.4	1.6	2.0	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.7	70.4	73.1	78.3	80.0	86.5	82.9	82.6	83.8	84.4	74.6	64.2	(19)
(20)			MCWB	55.1	59.6	56.0	59.9	61.4	66.2	65.8	65.7	66.4	62.8	58.6	55.2	(20)
(21)		2%	DB	62.2	65.8	67.7	72.3	73.3	77.4	76.1	76.7	78.5	78.1	69.5	61.5	(21)
(22)			MCWB	53.0	57.6	57.3	59.8	61.5	63.4	63.7	64.0	63.6	62.5	57.8	54.1	(22)
(23)		5%	DB	59.4	62.4	64.6	67.6	68.7	72.3	71.9	72.4	74.3	73.3	66.3	59.5	(23)
(24)			MCWB	53.0	55.7	56.2	57.6	59.7	61.4	62.2	62.8	63.1	60.6	55.8	53.6	(24)
(25)		10%	DB	57.2	59.8	61.8	64.4	65.4	68.8	69.0	69.6	70.9	69.7	63.2	58.0	(25)
(26)			MCWB	52.2	54.0	55.1	56.4	57.6	59.6	60.8	61.5	61.7	59.5	55.0	52.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.9	61.5	61.4	63.8	64.3	67.9	67.4	68.2	68.3	66.6	62.9	59.1	(27)
(28)			MCD B	60.7	68.8	69.7	75.0	74.0	85.6	83.0	78.3	78.4	81.6	68.0	61.4	(28)
(29)		2%	WB	56.1	58.9	59.2	61.4	62.4	65.2	65.3	66.1	65.5	64.2	60.1	57.2	(29)
(30)			MCD B	59.0	65.6	65.5	70.4	71.1	80.0	77.7	76.2	75.1	76.0	66.7	60.3	(30)
(31)		5%	WB	54.6	56.9	57.9	59.8	61.1	63.0	63.7	64.6	64.1	62.6	58.2	55.7	(31)
(32)			MCD B	57.8	62.5	63.3	67.2	69.1	74.5	73.0	73.7	73.2	72.2	65.0	59.0	(32)
(33)		10%	WB	52.9	55.5	56.8	58.4	59.5	61.3	62.3	63.3	62.6	61.2	56.7	53.8	(33)
(34)			MCD B	56.6	59.9	61.7	64.9	66.2	70.2	70.6	71.3	71.2	69.7	63.3	57.8	(34)
(35)	Mean Daily Temperature Range	MDBR		10.4	10.7	11.0	11.4	10.6	12.1	11.0	10.9	12.2	12.9	11.7	9.7	(35)
(36)		5% DB	MCD BR	13.9	14.9	15.3	17.9	16.9	19.5	16.3	16.2	18.8	19.7	16.2	11.1	(36)
(37)			MCWBR	8.6	8.9	8.5	9.3	8.8	10.4	8.2	8.9	9.5	9.7	9.6	8.3	(37)
(38)		5% WB	MCD BR	10.2	13.7	13.3	16.2	15.8	20.2	16.7	16.9	17.6	18.1	14.9	8.8	(38)
(39)			MCWBR	8.1	9.1	8.4	9.0	8.8	10.4	8.3	9.0	9.3	9.5	9.7	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.324	0.329	0.341	0.359	0.364	0.359	0.368	0.348	0.347	0.338	0.323	0.325	(40)
(41)		taud		2.555	2.537	2.491	2.408	2.423	2.442	2.419	2.475	2.463	2.529	2.556	2.556	(41)
(42)		Ebn at Noon		274	287	293	292	291	291	287	292	287	279	272	266	(42)
(43)		Edh at Noon		24	28	32	36	36	36	36	34	32	27	24	23	(43)
(44)	All-Sky Solar Radiation	RadAvg		754	1052	1488	1948	2245	2453	2321	2068	1753	1291	890	660	(44)
(45)		RadStd		106	129	134	154	145	105	79	90	67	83	58	97	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		N/A	N/A	-2.14	N/A	N/A	N/A	N/A	-198	+243	N/A

Nomenclature: See separate page